SETS, Relations and Functions - Second Edition

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Contents

Chapter 1

Preliminaries

1.1 Relations

Q: Show that the relation R in the set of real numbers defined as $R = \{(a,b) : \sqrt{a} + 3 \leqslant b\}$ is neither reflexive nor symmetric nor transitive.

1.1.1 Problems for Practice

1.1.1.1 Matrix match type problems

Matrix 1: Under column-I properties of relations are mentioned. Match the stated property with the relations listed under column-II , that satisfies it

Column I	Column II
(A) Equivalence	(P) The relation R in the set of natural numbers N, defined as $R = \{(a,b) : a > b^2\}$
(B) Reflexive	(Q) Relation R defined in the set $\{1,2,3,4,5\}$ defined by $R = \{(a,b) : a^2 + ab - 2b^2 = 0\}$
(C) Symmetric	(R) Relation R defined on the set of real numbers $\mathbb{R}$, defined by $R = \{(a,b) : \sqrt{a} + 5 < b\}$
(D) Transitive	(S) Relation R in the set $A = \{x \epsilon \mathbb{Z}; 0 \leq x \leq 9\}$,given by R=$\{(a,b) : (a-b)^4$ is divisible by 3$\}$

1.2 Real Numbers

When we talk of functions, we talk of only real valued functions. So, it is important that we brush up our theory of real numbers done in junior classes.

Definition

Any decimal fraction, terminating or nonterminating, is called a real number. The set of real numbers is represented by the symbol $\mathbb{R}$.

1.2.1 Rational Numbers

Every rational number can be written in the form $\frac{p}{q}$ where p and q are integers. The set of rational numbers is reqresented by the symbol $\mathbb{Q}$. i.e. $\mathbb{Q} = \{\frac{p}{q} \mid q \neq 0,\ p \ \& \ q \ \epsilon \ \mathbb{Z}\}$ where $\mathbb{Z}$ is the set of Integers.

The decimal representation of a rational number can be either terminating or non-terminating reccuring. e.g.

3.24 is a terminating decimal represenation.
$5.\overline{3}$ is a non-terminating recurring decimal representation.

Properties

- The set of rational numbers is **closed** with respect to the operations of addition, subtraction and product. i.e. If p and q are two rational numbers, then $p+q$, $p-q$ and pq are rational. Also $\frac{p}{q}$ is rational provided $q \neq 0$.**[Closure Property]**
- Between any two rational numbers p & q , there exist infinite rational numbers.[**Property of Denseness**]

Q. Find all the rational values of x for which $y = \sqrt{x^2+x+5}$ is a rational number.

Sol. Suppose x and y are rational numbers . Then the sum $x+y=q$ is a rational number.We now express x in terms of q .

$$y + x = \sqrt{x^2+x+5} + x = q$$

$$\sqrt{x^2+x+5} = q - x$$

$$x^2 + x + 5 = q^2 - 2qx + x^2$$

$$x = \frac{q^2-5}{1+2q}, q \neq -\frac{1}{2}$$

To prove the reverse i.e. $y = \sqrt{x^2+x+5}$ is a rational number if $x = \frac{q^2-5}{1+2q}, q \neq -\frac{1}{2}$

Its easy to check

$$y = \sqrt{x^2+x+5} = \sqrt{\left(\frac{q^2-5}{1+2q}\right)^2 + \left(\frac{q^2-5}{1+2q}\right) + 5}$$

$$= \sqrt{\frac{q^4+2q^3+11q^2+10q+25}{(1+2q)^2}}$$

$$= \sqrt{\left(\frac{q^2+q+5}{1+2q}\right)^2}$$

It may be observed that q^2+q+5 is positive for all values of q.Hence

$$y = \frac{q^2+q+5}{|1+2q|}$$

which is rational for $q \neq -\frac{1}{2}$.

Note : The sole purpose of adding x to y was to eliminate the square term of x . The same goal can be achieved by subtracting x from y.

1.2.2 Irrational Numbers

Non-terminating non-repeating decimal fractions are called Irrational numbers . In fact, all real numbers which are not rational are Irrational numbers. The set of Irrational numbers can be represented by the symbol T or $\mathbb{R} - \mathbb{Q}$ **.**

Properties

- The sum,difference,product and quotient of two Irrational numbers may be rational or irrational.ie. The set **T** is not closed with respect to addition,subtraction,multiplication and division.**[Closure Property]**
- The sum, difference,product and quotient of a rational number and an Irrational number is Irrational.
- There exist infinite rational and irrational numbers between two real numbers.**[Property of Denseness]**

Q. Prove that for every rational number r satisfying $r^2 < 3$ one can always find a larger rational number $r + k(k > 0)$for which $(r + k)^2 < 3$.

Sol. We know, there exist infinite rational numbers between a rational number and an irrational number. So, there will be infinite rational numbers between r and $\sqrt{3}$. So, there exist infinite values of k satisfying the condition.Let us take the case when r is sufficiently close to $\sqrt{3}$.In that case, we can chose a k such that $k \to 0$. This implies $k^2 < k$. i.e. $(r + k)^2 < r^2 + 2rk + k$. It is sufficient to put $r^2 + 2rk + k = 3$.i.e.

$$k = \frac{3 - r^2}{1 + 2r}$$

It may be observed that any rational number smaller than $\frac{3 - r^2}{1 + 2r}$ in magnitude will also satisfy the condition.

1.2.3 Problems for practice

1.2.3.1 Subjective Problems

Q1. Find all the rational values of x for which $y = \sqrt{49x^2 + 5x + 3}$ is a rational number.

Q2. Prove that for every rational number s satisfying $s^2 > 5$ one can always find a smaller rational number $s - k(k > 0)$for which $(s - k)^2 > 5$.

Q3. Does 219.1515515555155555555551......... represent a rational number?. Give reason.

Q4. Represent $\sqrt{31} - 3$ and $\sqrt{52}$ on the number line.

1.2.3.2 Multiple Answer MCQ's

Q1. Which of the following are rational numbers.

a) $\frac{\sqrt{3}}{2 + \sqrt{3}}$

b) $\frac{2}{6}$

c) $\left(\sqrt{5} - 2\right)\left(\sqrt{5} + 2\right)$

d) $\frac{11\pi}{7\pi}$

1.2.3.3 Write the Answer Type Questions

Q1: The solution set of $\dfrac{7}{x-2} \geq 5$ is

1.2.3.4 Assertion-Reason Type Questions

Each question in this section has four choices (a),(b),(c) and (d) out of which only one is correct. Mark your choices as follows:

(a) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is the correct explaination of STATEMENT-1.

(b) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is **NOT** the correct explaination of STATEMENT-1.

(c) STATEMENT-1 is True, STATEMENT-2 is False.

(d) STATEMENT-1 is False, STATEMENT-2 is True

Q1: STATEMENT-1: For every rational number r satisfying $r^2 < 2$ one can always find a larger rational number $r + k(k > 0)$for which $(r+k)^2 < 2$.

STATEMENT-2: There exist infinite rational numbers between any two rational numbers.

1.2.3.5 Hints and Solutions

Subjective Problems

Q1 $x = \dfrac{3-q^2}{14q-5}$, $q\epsilon\mathbb{Q}$ & $q \neq \dfrac{5}{14}$ {Hint: Put $y - 7x = q$ to get the answer. Also, equivalently if we had put $y + 7x = q$ we would have got , $x = \frac{q^2-3}{14q+5}$, $q \neq -\frac{5}{14}$}

Q2 {Hint: Let us take $s^2 - 2sk = 5$ which would imply $s^2 - 2sk + k^2 > 5$. ie $k = \frac{s^2-5}{2s}$}

Q3 No , it is not a rational number. {Hint. 2^{n-1}fives between nth and $(n+1)$th unity. The pattern is definite but non periodic.}

Q4 {Hint. $\sqrt{n}$ can be drawn by the following method of construction. Draw an arc from the origin of length $\frac{n-1}{2}$. From the point where it cuts the y axis, draw an arc of length $\frac{n+1}{2}$ intersecting the x-axis at P. The distance OP is $\sqrt{n}$.}

Multiple Answer MCQ's

Q1 B,C,D

Write the Answer Type Questions

Q1 the half open interval (2,17/5][1]

Assertion Reason Type Questions

Q1 B

1.3 Absolute value of a real number

The magnitude of a number is called the Absolute value of the number.Also,it is the distance of the point representing the number from origin on the number line . e.g. $|2-\sqrt{5}| = \sqrt{5} - 2$.

[1]Intervals can be of the following kinds

1. Open Interval i.e of the kind (a,b)
2. Closed Interval i.e. of the kind [a,b]
3. Semi-Closed Interval(or Semi-open/Half-open) ie. of the kind (a,b] or [a,b)

Mathematical Definition

$$|x| = \begin{cases} -x & ,x < 0 \\ x & ,x \geqslant 0 \end{cases}$$

Some alternate definitions of $|x|$ are

$$|x| = \begin{cases} \sqrt{x^2} \\ max\{x, -x\} \\ x.sgn(x) \end{cases}$$

1.3.1 Properties of Absolute value

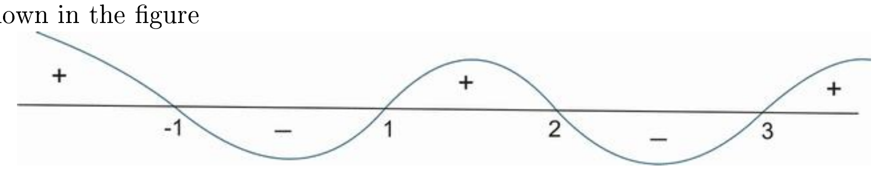
1. $|x| = \alpha(\alpha > 0) \Rightarrow x = \pm\alpha$
2. $|x| \leqslant \alpha(\alpha > 0) \Rightarrow -\alpha \leqslant x \leqslant \alpha$
3. $|x| \geqslant \alpha\,(\alpha > 0) \Rightarrow x \geqslant \alpha$or $x \leqslant -\alpha$
4. $|x+y| \leqslant |x| + |y|$ {Equality occurs when x and y are of same sign. i.e. $xy \geqslant 0$ e.g.$|\,(-5)+(-2)\,| = |-5| + |-2|$ as (-5).(-2) is positive }[**The Triangle Inequality**][2]
   - $|x-y| \leqslant |x| + |y|$ {Equality occurs when x and y are of opposite sign. i.e. $xy \leqslant 0$}
   - $|x+y| \geqslant ||x| - |y||$ {Equality occurs when x and y are of opposite sign. i.e. $xy \leqslant 0$}
   - $|x+y| \geqslant |x| - |y|$ {Equality occurs when x and y are of opposite sign i.e. $xy \leqslant 0$ and $|x| \geqslant |y|$}
   - $|x-y| \geqslant ||x| - |y||$ {Equality occurs when x and y are of same sign. i.e. $xy \geqslant 0$}
   - $|x-y| \geqslant |x| - |y|$ {Equality occurs when x and y are of same sign . i.e. $xy \geqslant 0$ and $|x| \geqslant |y|$}
5. $|xy| = |x||y|$
6. $\left|\dfrac{x}{y}\right| = \dfrac{|x|}{|y|}$,$y \neq 0$

1.3.2 The Wavey-Curve Technique

Let us consider, that we have an expression of the form

$$e = \frac{(x+1)\,(x-1)\,(x-3)}{x-2}$$

and we have to find the values of x for which this expression is > 0 , < 0 or $= 0$. To find the sign of the expression, we first of all mark these points on the number line . Then we draw a curve like the one shown in the figure



We start from the right hand side , draw a positive section of curve between ∞ and the right most point. Then, at the rightmost point(3 in this case) , we switch the sign of the curve to negative and

[2]A proof of the Triangle Inequality
$|x+y|^2 = (x+y)^2$
$= x^2 + 2xy + y^2$
$\leq x^2 + 2\,|x|\,|y| + y^2$
$= |x|^2 + 2\,|x|\,|y| + |y|^2$
$= (|x| + |y|)^2$
$\Longrightarrow |x+y| \leq$ |x|+|y|

take it below the axis. Like this , we change sign of the curve at every Critical point(ie. at 3,2,1,-1 in this case). Now from this curve, we try to find out the sign of the expression

$$e = \frac{(x+1)(x-1)(x-3)}{x-2}$$

$e > 0$ whenever $+$ sign appears in the wavey-curve(i.e. when wavey-curve is above the axis)
$\Rightarrow e > 0$ for $x \epsilon (-\infty, -1) \cup (1, 2) \cup (3, \infty)$[3]
$e < 0$ whenever - sign appears in the wavey-curve(i.e. when wavey-curve is below the axis)
$\Rightarrow e < 0$ for $x \epsilon (-1, 1) \cup (2, 3)$
$e = 0$ for all the critical points in the numerator(if and only if they don't appear in the denominator too)
$\Rightarrow e = 0$ for $x \epsilon \{-1, 1, 3\}$
$e =$Not defined for all the critical points in the denominator.
$e =$Not defined for $x \epsilon \{2\}$
{Note: It may be noted, that if we have to find x such that $e \geq 0$, then we will take the union of the values of x found out for $e > 0$ and $e = 0$
i.e. $e \geq 0$ for $x \epsilon ((-\infty, -1) \cup (1, 2) \cup (3, \infty)) \cup \{-1, 1, 3\}$. The same expression can be written by using a closed interval at the points where $e = 0$ i.e. $e \geq 0$ for $x \epsilon (-\infty, -1] \cup [1, 2) \cup [3, \infty)$
In a similar way, we can find the values of x for which $e \leq 0$. The solution is $x \epsilon [-1, 1] \cup (2, 3]$. }

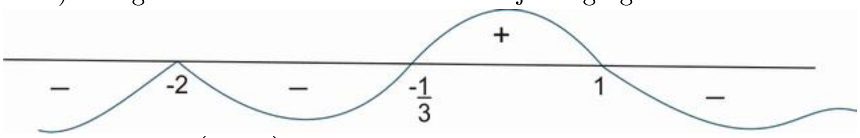
Example

We have to find the sign of $E = \dfrac{(x+2)^6 (3x+1)^3}{(1-x)}$

To find the sign of such an expression, we first of all make the coefficients of x as 1 in all terms. This gives

$$E = -\frac{27 (x+2)^6 \left(x + \frac{1}{3}\right)^3}{x-1}$$

The critical points of this expression are $x = \left\{-2, -\frac{1}{3}, 1\right\}$. The wavey-curve will start with a negative sign at ∞ as there is a - sign in front of E. The curve will then change sign at the first critical point -1. Thereafter , the curve will change sign 3 times at the point $x = -\frac{1}{3}$. (It may be noted that if a curve changes sign an odd number of times, it is equivalent to change of sign just one time) . Then the curve changes sign 6 times at $x = -2$.(change of sign an even number of times means the sign is retained).Things will be more clear from the adjoining figure.



$\Rightarrow E > 0$ for $x \epsilon \left(-\frac{1}{3}, 1\right)$

Similarly, $E < 0$ for $x \epsilon (-\infty, -2) \cup \left(-2, -\frac{1}{3}\right) \cup (1, \infty)$.

Also, $E = 0$ for $x \epsilon \left\{-2, -\frac{1}{3}\right\}$ and E is not defined for $x \epsilon \{1\}$

Q Solve for x

a) $|2x + 5| = 6$

[3]The actual reason for the positive sign of e in the interval $(3, \infty)$ is that in this interval the expressions $x - 3 > 0$, $x - 2 > 0$, $x - 1 > 0$ and also $x + 1 > 0$. Due to this reason, the expression e which is obtained by multiplying 3 positive quatities and dividing the resultant with a positive quantity is positive.
Similarly, in the interval $(2, 3)$ the quantities $x - 2 > 0$, $x - 1 > 0$ & $x + 1 > 0$ while the quantity $x - 3 < 0$. Hence the expression e is negative in this interval.
In this way we can investigate for other intervals also.

b) $|5x+7| = |7x-9\ |$

c) $|3x-5| < 1$

d) $|x^2-5x+6| > x^2-5x+6$

e) $|x| = x+7$

f) $|x| = x-5$

g) $\left|\dfrac{(x-5)^3}{(x+2)(x-3)}\right| > \dfrac{(x-5)^3}{(x+2)(x-3)}$

h) $|\cos x| = \cos x + 1$

i) $|(x^2+2x+5)+(3-5x)| = |x^2+2x+5| + |3-5x|$

j) $\left|\left(x^8-4\right)-\left(x^4+2\right)\right| = \left|x^8-4\right| - \left|x^4+2\right|$

k) $x^2-|x|-2=0$

l) $|x+2| = |x-2| + 2|x|$

m) $\left|\dfrac{3x+5}{5x+7}\right| - \left|\dfrac{7x+9}{5x+7}\right| = 1$

n)$2^{|x+1|} - |2^x-1| = 2^x+1$

Sol. a) $2x+5=\pm 6$

$\Rightarrow 2x = 1$ or $2x = -11$

$\Rightarrow x = \dfrac{1}{2}$ or $-\dfrac{11}{2}$

b) $|5x+7| = |7x-9\ |$

$\Rightarrow (5x+7) = \pm(7x-9)$

$\Rightarrow 5x+7 = 7x-9$ or $5x+7 = -(7x-9)$

$\Rightarrow 2x = 16$ or $12x = 2$

$\Rightarrow x = 8$ or $x = \dfrac{1}{6}$ i.e. $x\epsilon\left\{8, \dfrac{1}{6}\right\}$

c) **Method 1**: $|3x-5| < 1$

$\Rightarrow -1 < 3x-5 < 1$ (Property 2)

$\Rightarrow 4 < 3x < 6$

$\Rightarrow \dfrac{4}{3} < x < 2$ i.e $x\epsilon\left(\dfrac{4}{3}, 2\right)$

Method 2: $|3x-5| < 1$

Squaring both sides, we get

$(3x-5)^2 < 1$

$\Rightarrow 9x^2-30x+25 < 1$

$\Rightarrow 9x^2-30x+24 < 0$

$\Rightarrow 3x^2-10x+8 < 0$

$\Rightarrow (3x-4)(x-2) < 0$

$\Rightarrow x\epsilon\left(\dfrac{4}{3}, 2\right)$

d) $|x^2-5x+6| > x^2-5x+6$

{Note: It can be observed that $|y|$ is strictly greater than y if $y<0$}

$\Rightarrow x^2-5x+6<0$

$\Rightarrow (x-2)(x-3)<0$

$\Rightarrow 2<x<3$ i.e. $x\epsilon(2,3)$

e) **Case I**: $x\geq 0$

This assumption would imply that $|x|=x$

$\Rightarrow x=x+7$ which is not true for any x. Hence , no solution exists.

Case II: $x<0$

$\Rightarrow -x=x+7$

$\Rightarrow 2x=-7$ or $x=-\dfrac{7}{2}$ (which satisfies the condition $x<0$)

f)**Case I**: $x\geq 0$

$\Rightarrow x=x-5$ which is not true for any x. Hence , no solution exists for $x\geq 0$.

Case II: $x<0$

$\Rightarrow -x=x-5$

$\Rightarrow 2x=5$ or $x=\dfrac{5}{2}$

This value of x does not satisfy $x<0$. Hence , no solution exists for $x<0$ also.

g)$\left|\dfrac{(x-5)^3}{(x+2)^2(x-3)}\right|>\dfrac{(x-5)^3}{(x+2)^2(x-3)}$

It may be observed that $|y|>y$ is only possible when $y<0$

$\Rightarrow \dfrac{(x-5)^3}{(x+2)^2(x-3)}<0$

$\Rightarrow x\epsilon(3,5)$

[Note:The solution to such equations can be found out with the help of **Wavey-Curve Technique** as discussed in the Lectures]

h) $|\cos x|=\cos x+1$

Case I: $\cos x\geq 0$

$\Rightarrow \cos x=\cos x+1$

which is not possible for any x. Hence , no solution exists for $\cos x\geq 0$.

Case II:$\cos x<0$

$\Rightarrow -\cos x=\cos x+1$

$\Rightarrow 2\cos x=-1$

$\Rightarrow \cos x=-\dfrac{1}{2}$ (which satisies our assumption that $\cos x<0$)

$\Rightarrow x=2n\pi\pm\dfrac{2\pi}{3}$, $n\epsilon Z$.

i) $|(x^2+2x+5)+(3-5x)|=|x^2+2x+5|+|3-5x|$

$|x+y|=|x|+|y|\Rightarrow xy\geq 0$(equality condition in Property 4)

i.e x & y are of the same sign

Now $x^2+2x+5=(x+1)^2+4>0$

$\Rightarrow 3-5x\geq 0$

i.e $x\leq\dfrac{3}{5}$ or $x\epsilon$(-$\infty,\dfrac{3}{5}$)

j)$\left|(x^8-4)-(x^4+2)\right|=\left|x^8-4\right|-\left|x^4+2\right|$

We know, $|x-y|=|x|-|y|$ if $xy \geq 0$ & $|x| \geqslant |y|$

Also , it may be noted that $x^4+4>0 \forall x$

$\Rightarrow$Our conditions reduce to $x^8-4 \geq x^4+2$

$\Rightarrow x^4-2 \geq 1$

$\Rightarrow x^4 \geq 3$

$\Rightarrow x \geq \sqrt[4]{3}$ or $x \leq -\sqrt[4]{3}$ i.e. $x \epsilon \left(-\infty,-\sqrt[4]{3}\right) \cup \left(\sqrt[4]{3},\infty\right)$

k)$x^2-|x|-2=0$

$\Rightarrow |x|^2-|x|-2=0$

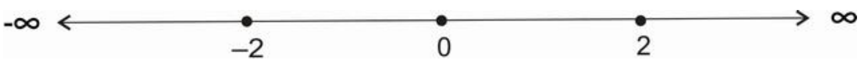
$\Rightarrow (|x|-2)(|x|+1)=0$

$\Rightarrow |x|=2$ or $|x|=-1$(Rejected)

$\Rightarrow x=\pm 2$ i.e $x \epsilon \{-2,2\}$

l)$|x+2|=|x-2|+2|x|$

We first of all find the critical points. $|x+2|=0$ gives the point -2. Similarly, the points 2,0 are obtained by euating the other modulus terms to zero. So, there are 3 critical points ie at $x=\{-2,0,2\}$. We plot these points on the number line and divide the whole number line at these critical points.



Case I : $x<-2$

The equation becomes

$-(x+2)=-(x-2)-2x$

$\Rightarrow 2x=4$

$\Rightarrow x=2$ (which is rejected as it does not belong to the interval $(-\infty,-2)$)

Case II: $-2 \leq x<0$

The equation becomes

$x+2=-(x-2)-2x$

$\Rightarrow x=0$ (Which is rejected as it does not belong to the interval $[-2,0)$)

Case III : $0 \leq x<2$

The equation becomes

$x+2=-(x-2)+2x$

which is true for all x , but our interval is $[0,2)$

$\Rightarrow x \epsilon [0,2)$ is a solution

Case IV : $x \geq 2$

The equation in this case becomes

$x+2=(x-2)+2x$

$\Rightarrow x=2$ (which is accepted as it lies in the interval chosen i.e. $[2,\infty)$)

Hence $x=2$ is a solution

Now we combine all the solutions obtained from different intervals. So, the final solution is the union of all the cases i.e. $x \epsilon [0,2) \cup \{2\}$

$\Rightarrow$The complete solution set is $x \epsilon [0,2]$

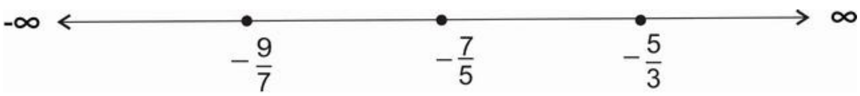
m)$\left|\dfrac{3x+5}{5x+7}\right|-\left|\dfrac{7x+9}{5x+7}\right|=1$

First thing to be noted is that $5x+7 \neq 0$

i.e. $x \neq -\frac{7}{5}$

$\Rightarrow |3x+5| - |7x+9| = |5x+7|$

Now in this case, the critical points are $-\frac{9}{7}$, $-\frac{7}{5}$ and $-\frac{5}{3}$. Plotting them on the number line, we get the intervals as shown below



Case I: $x < -\frac{9}{7}$

$\Rightarrow -(3x+5) + (7x+9) = -(5x+7)$

$\Rightarrow 9x = -11$

$\Rightarrow x = -\frac{11}{9}$ (which lies in the interval defined by $x < -\frac{9}{7}$)

Case II: $-\frac{9}{7} \leq x < -\frac{7}{5}$

$\Rightarrow -(3x+5) - (7x+9) = -(5x+7)$

$\Rightarrow -5x = 7$

$\Rightarrow x = -\frac{7}{5}$ (which is rejected because in the beginning we fixed $x \neq -\frac{7}{5}$. Also the point $-\frac{7}{5}$ does not lie in the interval $-\frac{9}{7} \leq x < -\frac{7}{5}$)

Case III : $-\frac{7}{5} \leq x < -\frac{5}{3}$

$\Rightarrow -(3x+5) - (7x+9) = +(5x+7)$

$\Rightarrow -15x = 21$

$\Rightarrow x = -\frac{7}{5}$ (which is rejected because in the beginning we fixed $x \neq -\frac{7}{5}$ due to domain restrictions.)

Case IV: $x \geq -\frac{5}{3}$

$\Rightarrow (3x+5) - (7x+9) = (5x+7)$

$\Rightarrow -9x = 11$

$\Rightarrow x = -\frac{11}{9}$ (Rejected because $-\frac{11}{9}$ does not belong to the interval $x \geq -\frac{5}{3}$)

Hence, the only solution is $x = -\frac{11}{9}$.

n)$2^{|x+1|} - |2^x - 1| = 2^x + 1$

The critical points lie at $|x+1| = 0$ and $|2^x - 1| = 0$. i.e at $x = -1$ and $x = 0$ respectively. So , dividing the number line as done previously

Case I: $x < -1$

$\Rightarrow 2^{-(x+1)} + (2^x - 1) = 2^x + 1$

$\Rightarrow 2^{-(x+1)} = 2$

$\Rightarrow -(x+1) = 1$

$\Rightarrow x = -2$ (Accepted)

Case II: $-1 \leq x < 0$

$\Rightarrow 2^{x+1} + 2^x - 1 = 2^x + 1$

$\Rightarrow 2^{x+1} = 2$

$\Rightarrow x + 1 = 1$

$\Rightarrow x = 0$ (Rejected as it does not lie in the interval $-1 \leq x < 0$)

Case III: $x \geq 0$

$\Rightarrow 2^{x+1} - (2^x - 1) = 2^x + 1$

$\Rightarrow 2^{x+1} = 2.2^x$ (which is true for all x belonging to the interval $x \geq 0$)

$\Rightarrow$The solution to Case III is $x \epsilon [0, \infty)$

Hence, the general solution to this problem is $x \epsilon \{-2\} \cup [0, \infty)$.

1.3.3 Problems for Practice

1.3.3.1 Subjective Problems

Q1: Solve for x

a) $|\sin x| = -\sin x + 2$

b) $\left|\dfrac{(x+3)^4 (x-3)^3}{(x+1)(x-1)}\right| < -\dfrac{(x+3)^4 (x-3)^3}{(x+1)(x-1)}$

c) $\left|x^2 + 3x + 2\right| = -\left(x^2 + 3x + 2\right)$

1.3.3.2 Single Answer MCQ's

Q1: The complete solution set of $\dfrac{1}{\left|\dfrac{4}{x} - 10\right|} < 3$ is

a) $x \epsilon \mathbb{R} - \left(\dfrac{12}{31}, \dfrac{12}{29}\right)$

b) $x \epsilon \left(\dfrac{12}{31}, \dfrac{12}{29}\right) - \left\{\dfrac{2}{5}\right\}$

c) $x \epsilon \mathbb{R} - \{0\} - \left[\dfrac{12}{31}, \dfrac{12}{29}\right]$

d) None of these

Q2: The point in the co-ordinate plane satisfying $|7x + 5| = -|3 - 3y|$ is

a)$(-2, 3)$

b)$\left(-\dfrac{5}{7}, 1\right)$

c) $(-5, -3)$

d) None of these

Q3: The complete solution set of $|\cot x| = \cot x + 2\sqrt{3}$ is

a)$\{x : x = n\pi + \dfrac{\pi}{3}\ , n\epsilon\mathbb{Z}\}$

b)$\{x : x = n\pi - \dfrac{\pi}{3}\ , n\epsilon\mathbb{Z}\}$

c)$\{x : x = n\pi + \dfrac{\pi}{6}\ , n\epsilon\mathbb{Z}\}$

d)$\{x : x = n\pi - \dfrac{\pi}{6}\ , n\epsilon\mathbb{Z}\}$

Q4: The area bounded by the curves given by $|x| + |2y| = 2$ is

a) 2 units

b) 4 units

c) 6 units

d) 8 units

1.3.3.3 Multiple answer MCQ's

Q1: The solution set of $2|\tan x| = \tan x + \sqrt{3}$ contains

a)$\{x : x = n\pi + \frac{\pi}{3} \text{ , } n\epsilon Z\}$

b)$\{x : x = n\pi - \frac{\pi}{3} \text{ , } n\epsilon Z\}$

c)$\{x : x = n\pi + \frac{\pi}{6} \text{ , } n\epsilon Z\}$

d)$\{x : x = n\pi - \frac{\pi}{6} \text{ , } n\epsilon Z\}$

1.3.3.4 Matrix Match Type Problems

Matrix 1: Under Column I , some equations are given . Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II.{Note: [ ] is the Greatest Integer Function}

Column I	**Column II**
(P)$\left\|\sin^{-1} x\right\| = \sin^{-1} x + \frac{\pi}{6}$	**(A)**-$\frac{1}{2}$
(Q)$\ln\|x\| = \|\ln x\|$	**(B)**1
(R)$\|[x]\| = [\|x\|]$	**(C)**$\frac{3}{2}$
(S)$\left\|x^2 - 3x + 2\right\| > x^2 - 3x + 2$	**(D)**2

1.3.3.5 Linked Comprehension Type Problems

Comprehension 1: Consider the equation $y = \left|\left(x^4 - 9\right) - \left(x^2 - 3\right)\right| - \left|x^4 - 9\right| + \left|x^2 - 3\right|$. The sign of y depends on the value of x. Read the following questions based on this information and answer them carefully.

Q1: The value(s) of x for which $y > 0$ is/are :

a)$x\epsilon R$

b)$x\epsilon\left(-\sqrt{3}, \sqrt{3}\right)$

c)$x\epsilon\left(-\sqrt[4]{3}, \sqrt{3}\right)$

d)None of these

Q2: The value(s) of x for which $y \leq 0$ is/are :

a)$x\epsilon R$

b)$x\epsilon\left(-\sqrt{3}, \sqrt{3}\right)$

c)$x\epsilon\left(-\sqrt[4]{3}, \sqrt{3}\right)$

d)None of these

Comprehension 2: The first 3 terms of an A.P. are $|3x|$, $|3x - 1|$and $|3x + 1|$. Based on this information, give the answers to the questions below

Q1: The value of x is

a) 1

b) 2

c) -1

d) None of these

Q2: The 10th term of the A.P. is

a) $\frac{1}{4}$

b) $\frac{5}{4}$

c) $\frac{19}{4}$

d) None of these

Q3: The sum of first 15 terms of the series is

a) $\frac{225}{4}$

b) $\frac{359}{4}$

c) $\frac{433}{4}$

d) None of these

Comprehension 3: A function $f_n(x)$ is defined for all $n \epsilon N$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x \epsilon R - \{-1\}$. Using this definition , $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x \epsilon R - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x \epsilon R - \left\{-1, 0, \frac{1}{2}\right\}$and so on. Based on the information, answer the questions below.

Q1: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

a) $(-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}$

b) $(-\infty, 1] - \left\{-1, 0, \frac{1}{2}\right\}$

c) $(1, \infty) - \{2\}$

d) None of these

Q2: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ is

a) $(-1, 1)$

b) $\mathbb{R} - \{2\} - [-1, 1]$

c) $\mathbb{R} - \{-1, 1, 2\}$

d) None of these

Q3: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

a) $\left(\frac{1}{2}, 2\right)$

b) $(2, \infty)$

c) $(0, 1)$

d) None of these

1.3.3.6 Hints and Solutions

Subjective Questions

Q1 a)$x = 2n\pi + \dfrac{\pi}{2}$, $n\epsilon Z$

{Hint: The solution of the equation is all values of x which satisfy the equation , $\sin x = 1$}

b) No Solution i.e $x\epsilon\phi$

{Hint: It may be observed that $|y| < -y$ is never true . Even $|y| < y$ also gives no solution.}

c) $x\epsilon\,[-3, -2]$

{Hint: $|y| = -y \Rightarrow y \leq 0$ }

Single Answer MCQ's

Q1) C

{Hint: $\dfrac{1}{\left|\dfrac{4}{x} - 10\right|} < 3$

$\Rightarrow \left|\dfrac{4}{x} - 10\right| > \dfrac{1}{3}$

Case I : $\dfrac{4}{x} - 10 > \dfrac{1}{3}$

$\Rightarrow \dfrac{4}{x} > \dfrac{31}{3}$

$\Rightarrow \dfrac{1}{x} > \dfrac{31}{12}$

$\Rightarrow 0 < x < \dfrac{12}{31}$

Case II: $\dfrac{4}{x} - 10 < -\dfrac{1}{3}$

$\Rightarrow \dfrac{4}{x} < \dfrac{29}{3}$

$\Rightarrow \dfrac{1}{x} < \dfrac{29}{12}$

$\Rightarrow \dfrac{1}{x} < 0$ or $0 < \dfrac{1}{x} < \dfrac{29}{12}$

$\Rightarrow x < 0$ or $x > \dfrac{12}{29}$

Combining all cases, we get $x\epsilon\,(-\infty, 0) \cup \left(0, \dfrac{12}{31}\right) \cup \left(\dfrac{12}{29}, \infty\right)$ i.e. $x\epsilon\mathbb{R} - \{0\} - \left[\dfrac{12}{31}, \dfrac{12}{29}\right]$}

Q2) B

Q3) D

Q4) B

Multiple Answer MCQ's

Q1) A,D

Matrix Match Type Problems

Matrix 1:

$P \rightarrow A$

$Q \rightarrow B, C, D$

$R \rightarrow B, C, D$

$S \rightarrow C$

Linked Comprehension Type Problems

Comprehension 1: Q1) D Q2)A

{Hint:a) $y > 0$

$\Rightarrow \left|(x^4 - 9) - (x^2 - 3)\right| - |x^4 - 9| + |x^2 - 3| > 0$

$\Rightarrow \left|\left(x^4-9\right)-\left(x^2-3\right)\right| > \left|x^4-9\right|-\left|x^2-3\right|$

Now we know that $|x-y| \geqslant |x|-|y|$ {Equality occurs when x and y are of same sign . i.e. $xy \geqslant 0$ and $|x| \geqslant |y|$}

$\Rightarrow \left|\left(x^4-9\right)-\left(x^2-3\right)\right| = \left|x^4-9\right|-\left|x^2-3\right|$ when $\left(x^4-9\right)\left(x^2-3\right) \geqslant 0$ and $\left|x^4-9\right| \geqslant \left|x^2-3\right|$

i.e when $\left(x^2-3\right)^2\left(x^2+3\right) \geqslant 0$ and $\left|x^2-3\right|\left(x^2+3\right) \geqslant \left|x^2-3\right|$

i.e when $\left(x^2-3\right)^2\left(x^2+3\right) \geqslant 0$ and $\left|x^2-3\right|\left(x^2+2\right) \geqslant 0$ which is true for all x.

i.e $\forall x \epsilon \mathbb{R}$, y is identically equal to zero.

Hence $y > 0$ has no solution

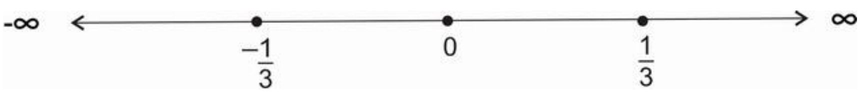
and $y \leqslant 0$ is true for all $x\epsilon\mathbb{R}$. i.e (b)$\rightarrow$A}

Comprehension 2: Q1) D Q2)C Q3)A

{Hint: a)$|3x|$, $|3x-1|$and $|3x+1|$are in A.P.

$\Rightarrow 2 \times |3x-1| = |3x| + |3x+1|$

The critical points in this case are $\left\{-\frac{1}{3}, 0, \frac{1}{3}\right\}$



Case I : $x\epsilon\left(-\infty, -\frac{1}{3}\right)$

In this case equation becomes

$2 \times -(3x-1) = -3x-(3x+1)$

$\Rightarrow -6x+2 = -6x-1$ which gives no solution

Case II: $x\epsilon\left[-\frac{1}{3}, 0\right)$

In this case , the equation becomes

$2 \times (3x-1) = -3x-(3x+1)$

$\Rightarrow 6x-2 = -6x-1$

$\Rightarrow x = \frac{1}{12}$ (which is accepted as $\frac{1}{12}\epsilon\left[-\frac{1}{3}, 0\right)$)

Case III : $x\epsilon\left[0, \frac{1}{3}\right)$

In this case , the equation becomes

$2 \times (3x-1) = 3x-(3x+1)$

$\Rightarrow 6x-2 = -1$

$\Rightarrow x = \frac{1}{6}$ (which is rejected as $\frac{1}{6} \notin \left[0, \frac{1}{3}\right)$

Case IV :$x\epsilon\left[\frac{1}{3}, \infty\right)$

In this case, the equation becomes

$2 \times (3x-1) = 3x+(3x+1)$

i.e $6x-2 = 6x+1$. Hence no solution in this case also.

Combining all the cases, we get $x = \frac{1}{12}$

b) Substituting the value of x , the series becomes $\left\{\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \cdots\cdots\cdots\right\}$

Hence, the series has first term $a = \frac{1}{4}$ and common difference $d = \frac{1}{2}$.

$T_{10} = a + (10-1)\,d = \frac{1}{4} + 9 \times \frac{1}{2} = \frac{19}{4}$

c) The sum of first 15 terms of the series is

$S_{15} = \frac{n}{2}\left(2a+(n-1)\,d\right) = \frac{15}{2}\left(2 \times \frac{1}{4} + 9 \times \frac{1}{2}\right) = \frac{15}{2} \times \frac{15}{2} = \frac{225}{4}$}

Comprehension 3: Q1) A Q2)B Q3)C

{Hint: $f_1(x) = \dfrac{2x-1}{x+1}$ for $x\epsilon R - \{-1\}$

$f_2(x) = f_1(f_1(x)) = \dfrac{x-1}{x}$ for $x\epsilon R - \{-1, 0\}$

$f_3(x) = f_1(f_2(x)) = \dfrac{2-x}{1-2x}$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}\right\}$

On simlar lines, we can find $f_4(x) = f_1(f_3(x)) = \dfrac{1}{1-x}$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1\right\}$

$f_5(x) = f_1(f_4(x)) = \dfrac{1+x}{2-x}$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$f_6(x) = f_1(f_5(x)) = x$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

It may be noted that this function repeats itself after a gap of 6 .i.e $f_{6n+r}(x) = f_6(f_6 ... ntimes(f_r(x)))) = f_r(x)$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

Q1: $f_{34}(x) = f_4(x)$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$\Rightarrow f_{34}(x) = \dfrac{1}{1-x}$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$\Rightarrow g(x) = |\ln(f_{34}(x))|$ has domain given by

$\dfrac{1}{1-x} > 0$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

i.e $x < 1$ and $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

i.e $x\epsilon \left((-\infty, 1) - \left\{-1, 0, \dfrac{1}{2}\right\}\right)$

Q2: $|f_{71}\,(x)| > \left|\dfrac{1}{f_{73}\,(x)}\right|$

i.e we have to solve $|f_5\,(x)| > \left|\dfrac{1}{f_1\,(x)}\right|$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

i.e $\left|\dfrac{1+x}{2-x}\right| > \left|\dfrac{1}{\dfrac{2x-1}{x+1}}\right|$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

i.e $|2x-1| > |2-x|$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

Squaring both sides we get

$4x^2 - 4x + 1 > 4 + x^2 - 4x$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

i.e. $3x^2 > 3$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$\Rightarrow x\epsilon\,(\mathbb{R} - \{2\} - [-1, 1])$

Q3: $|f_{56}\,(x)| > f_{56}\,(x)$

$\Rightarrow |f_2(x)| > f_2(x)$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$\Rightarrow f_2(x) < 0$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$\Rightarrow \dfrac{x-1}{x} < 0$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}, 1, 2\right\}$

$\Rightarrow x\epsilon \left((0, 1) - \left\{\dfrac{1}{2}\right\}\right)$

These values are a substet of $(0, 1)$.}

Chapter 2

Functions (Preliminaries II)

Functions are one of the most important areas of mathematics because they lie at the heart of much of mathematical analysis.

2.1 Domain of Definition and Range

Q: For the function $f(x) = \dfrac{x}{x+2}$, find $f(x+3)$, $f(3x)$,$3f(x)$, $3f(3x+3)$, $f\left(x^3\right)$, $(f(x))^3$ with their respective domains of definition.

Sol: $f(x+3) = \dfrac{x+3}{x+5}$ where $x+5 \neq 0$ i.e $D_{f(x+3)} = \mathbb{R} - \{-5\}$

$f(3x) = \dfrac{3x}{3x+2}$ where $3x+2 \neq 0$ i.e. $D_{f(3x)} = \mathbb{R} - \left\{-\dfrac{2}{3}\right\}$

$3f(x) = \dfrac{3x}{x+2}$ where $x+2 \neq 0$ i.e. $D_{3f(x)} = \mathbb{R} - \{-2\}$

$3f(3x+3) = \dfrac{3x}{3x+5}$ where $3x+5 \neq 0$ i.e. $D_{3f(3x+3)} = \mathbb{R} - \left\{-\dfrac{5}{3}\right\}$

$f\left(x^3\right) = \dfrac{x^3}{x^3+2}$ where $x^3+2 \neq 0$ i.e. $D_{f\left(x^3\right)} = \mathbb{R} - \{-\sqrt[3]{2}\}$

$(f(x))^3 = \left(\dfrac{x}{x+2}\right)^3$ where $x+2 \neq 0$ i.e. $D_{(f(x))^3} = \mathbb{R} - \{-2\}$

Q: Find the domains of definition of the following functions:

a) $\sqrt{2-x} - \sqrt[3]{x} + \sqrt[4]{2+x}$

b) $\sqrt{x^2+x-2}$

c) $\dfrac{\sqrt{x^2-3x+2}}{\sqrt{4-x^2}}$

d) $\sqrt{\tan x}$

e) $\sqrt{-\sin x - \dfrac{1}{2}}$

f) $\sqrt{1 - cosecx}$

g) $\log\left(\dfrac{x^3\left(x^2-4\right)}{\left(x^2-1\right)(x+3)}\right)$

h) $\sqrt{\log x}$

i) $\sqrt{\log_x(2-x)}$

j) $\cos^{-1}\left(\frac{x}{2}-2\right)-\log\left(\frac{x}{2}-2\right)$

k) $\sqrt{[x]-x}$ where [ ] represents the greatest integer function

l) $\sqrt{\sin\left(\frac{[x]}{2}\pi\right)}$

m) $\frac{1}{\sqrt{|x|-x}}$

2.1.1 Problems for Practice

2.1.1.1 Subjective Problems

Q1: (i) Prove that for the logarithmic function $y=\ln|x|$, if the argument takes values in a G.P. , then the corresponding values of the function y are in A.P.

(ii)Prove that for the exponential function $y=e^{x}$, if the argument takes values in a A.P.. , then the corresponding values of the function y are in G.P.

2.1.1.2 Hints and Solutions

Subjective Problems

Q1 : {Hint: (i) Let the arguments be defined by the general term $x_n=ar^{n-1}$. Then the value of the function corresponding to it will be $y_n=\ln\left|ar^{n-1}\right|=\ln|a|+(n-1)\ln|r|$. This means that two consecutive terms y_n and y_{n-1} differ by a constant .i.e. $y_n-y_{n-1}=\ln|r|$. Hence , these terms are in A.P.

(ii)Proceeding in a similar manner , if we take the argument as a general term of an A.P. then the corresponding values of the function will have a constant common difference.}

2.2 Periodicity

2.2.1 Problems for practice

2.2.1.1 Subjective Problems

Q1: Find the range of the function defined on **R** given by

$$f(x)=\begin{cases} e^{x-n} & ;n\leqslant x\leqslant n+\frac{2}{3} \\ \sin\left(\frac{3\pi}{2}x\right) & ;n+\frac{2}{3}<x<n+1 \end{cases}$$

where $n\epsilon Z$.

2.3 Injectivity, Surjectivity and Bijectivity

2.3.1 Problems for Practice

2.3.1.1 Subjective Problems

Q1: Check whether the function $f:(-\infty,-3)\to\left[-\frac{1}{4},0\right]$ given by $f(x)=\frac{x+1}{x^2+2x+5}$ is bijective in its domain or not. Given that $f(x)$ is strictly decreasing in the domain.

2.3.1.2 Single Answer MCQ's

Q1: A function $f:R\to[1,2]$ given by $f(x)=\log_3(9-6|\cos x|)$ is

a) One-one only

b) Onto only

c) One-one and onto

d) None of these

Q2: Let $f:[0,\infty)\to S$ defined as $f(x)=2^{x(1-x)}$ be onto, then the set S is

a) $[0,\infty)$

b) $[1,\infty)$

c) $\left[2^{-\frac{1}{4}},1\right]$

d) $\left[2^{-\frac{1}{4}},\infty\right)$

2.3.1.3 Hints and Solutions

Single Answer MCQ's

Q1: B

Q2: D

2.4 Inverse of a function

2.4.1 Problems for practice

2.4.1.1 Subjective Problems

Q1: Check whether the function $f:(-\infty,0)\to(0,\infty)$ given by $f(x)=\dfrac{1}{\sqrt{|x|-x}}$ is one-one and onto. Find it's inverse if possible.

Q2: A function $f:R-\{-2,2\}\to\left(-\infty,\dfrac{5}{2}\right]\cup(3,\infty)$ given by $f(x)=3+\dfrac{1}{|x|-2}$. Divide the domain into minimum number of suitable subsets so that the function becomes invertible on each individual domain. Also find the inverse on each of these domains.

2.4.1.2 Single Answer MCQ's

Q1: A function on $f:[2,4)\to(-\infty,0]$ is given by $f(x)=-2+\log_2\left(-x^2+4x\right)$. Then $f^{-1}(x)$ is given by

a) $2\pm2\sqrt{1-2^x}$

b) $2-2\sqrt{1-2^x}$

c)$2+2\sqrt{4-2^x}$

d) None of these

2.4.1.3 Hints and Solutions

Single Answer MCQ's

Q1: D